EXPERIMENT NO: 6

ONLINE SHOPPING APPLICATION - POSITIVE TEST SCENARIOS

Objective: To analyze the requirements of an online shopping application and prepare positive test scenarios and test cases.

Case Study: The online shopping application enables customers to sign in, browse products, filter items, add products to a cart, apply discounts, complete payment, track shipments and submit reviews.

Requirement Analysis:

- REQ-01: Customers shall sign in using valid registered credentials.
- REQ-02: Customers shall search and filter products by category or keyword.
- REQ-03: Customers shall add available products to the cart and update quantities.
- REQ-04: Valid discount codes shall update the order total.
- REQ-05: Successful payment shall create an order and shipment reference.

Features to be tested:

1. Customer sign-in
2. Product search
3. Cart management
4. Discount code
5. Checkout and payment
6. Shipment tracking
7. Product review

Test Case ID: SHOP_P01
Test Description: Sign in with valid credentials

Steps to reproduce:

1. 1. Open the shopping website.
2. 2. Enter a registered email and correct password.

3. 3. Select Sign In.

Expected Result: The customer dashboard is displayed.

Actual Result: The customer dashboard is displayed.

Status: Pass

Test Case ID: SHOP_P02
Test Description: Search for a product by keyword

Steps to reproduce:

4. 1. Open the product search field.
5. 2. Enter 'wireless headphones'.
6. 3. Select Search.

Expected Result: Matching available products are listed.

Actual Result: Matching available products are listed.

Status: Pass

Test Case ID: SHOP_P03
Test Description: Add a product to the cart

Steps to reproduce:

7. 1. Open a product page.
8. 2. Choose an available colour or size if required.
9. 3. Select Add to Cart and open the cart.

Expected Result: The selected product, quantity and price appear correctly in the cart.

Actual Result: The selected product, quantity and price appear correctly in the cart.

Status: Pass

Test Case ID: SHOP_P04
Test Description: Apply a valid discount code

Steps to reproduce:

10.1. Open the cart.
11.2. Enter a valid discount code.

12.3. Select Apply.

Expected Result: The discount is applied and the final amount is recalculated.

Actual Result: The discount is applied and the final amount is recalculated.

Status: Pass

Test Case ID: SHOP_P05
Test Description: Complete checkout using a valid payment method

Steps to reproduce:

13.1. Enter a valid delivery address.
14.2. Choose an available payment method.
15.3. Complete the payment.

Expected Result: An order confirmation and order number are displayed.

Actual Result: An order confirmation and order number are displayed.

Status: Pass

Test Case ID: SHOP_P06
Test Description: View shipment tracking

Steps to reproduce:

16.1. Open My Orders.
17.2. Select an order that has shipped.

Expected Result: The current shipment status and estimated delivery information are displayed.

Actual Result: The current shipment status and estimated delivery information are displayed.

Status: Pass

Test Case ID: SHOP_P07
Test Description: Submit a review for a delivered product

Steps to reproduce:

18.1. Open a delivered order.

19.2. Select a star rating and enter a review.
20.3. Select Submit Review.

Expected Result: The review confirmation is displayed.

Actual Result: The review confirmation is displayed.

Status: Pass

Result: 7 of 7 test cases passed.

EXPERIMENT NO: 7

ONLINE SHOPPING APPLICATION - NEGATIVE TEST SCENARIOS

Objective: To verify that the shopping application rejects invalid input and prevents incorrect orders or payments.

Case Study: Negative testing checks invalid credentials, unavailable products, excessive quantities, expired coupons, invalid addresses and failed payments. The application should display clear messages and preserve order and payment integrity.

Requirement Analysis:

- REQ-01: Incorrect credentials shall not grant access.
- REQ-02: Out-of-stock products shall not be ordered.
- REQ-03: Quantity shall remain within the configured limit.
- REQ-04: Expired or invalid discount codes shall be rejected.
- REQ-05: Failed payments shall not create confirmed orders.

Features to be tested:

1. Login validation

2. Stock validation

3. Quantity limit

4. Coupon validation

5. Address validation

6. Payment failure

7. Duplicate submission

Test Case ID: SHOP_N01
Test Description: Sign in with an incorrect password

Steps to reproduce:

21.1. Open the sign-in page.
22.2. Enter a registered email and an incorrect password.
23.3. Select Sign In.

Expected Result: An invalid-credentials message is displayed and access is denied.

Actual Result: An invalid-credentials message is displayed and access is denied.

Status: Pass

Test Case ID: SHOP_N02
Test Description: Attempt to order an out-of-stock product

Steps to reproduce:

24.1. Open a product marked Out of Stock.
25.2. Try to add it to the cart.

Expected Result: The product cannot be added and an availability message is shown.

Actual Result: The product cannot be added and an availability message is shown.

Status: Pass

Test Case ID: SHOP_N03
Test Description: Enter a quantity above the allowed limit

Steps to reproduce:

26.1. Open an available product.
27.2. Set the quantity above the configured maximum.
28.3. Attempt to update the cart.

Expected Result: The quantity is rejected and the maximum allowed quantity is explained.

Actual Result: The quantity is rejected and the maximum allowed quantity is explained.

Status: Pass

Test Case ID: SHOP_N04
Test Description: Apply an expired discount code

Steps to reproduce:

29.1. Open the cart.
30.2. Enter an expired code.
31.3. Select Apply.

Expected Result: An expiry message appears and the total remains unchanged.

Actual Result: An expiry message appears and the total remains unchanged.

Status: Pass

Test Case ID: SHOP_N05
Test Description: Submit checkout with an incomplete address

Steps to reproduce:

32.1. Proceed to checkout.
33.2. Leave a required address field blank.
34.3. Select Continue.

Expected Result: Field-level validation is displayed and the order is not submitted.

Actual Result: Field-level validation is displayed and the order is not submitted.

Status: Pass

Test Case ID: SHOP_N06
Test Description: Attempt checkout after payment failure

Steps to reproduce:

35.1. Proceed to payment.
36.2. Use a declined test payment method.
37.3. Return to the order page.

Expected Result: Payment failure is shown and no confirmed order or duplicate charge is created.

Actual Result: Payment failure is shown and no confirmed order or duplicate charge is created.

Status: Pass

Test Case ID: SHOP_N07
Test Description: Click the payment button repeatedly

Steps to reproduce:

38.1. Proceed to payment with valid details.
39.2. Click Pay repeatedly while the first request is processing.

Expected Result: Only one order/payment request is processed and duplicate orders are prevented.

Actual Result: Only one order/payment request is processed and duplicate orders are prevented.

Status: Pass

Result: 7 of 7 test cases passed.

EXPERIMENT NO: 8

HOSPITAL APPOINTMENT SYSTEM - POSITIVE TEST SCENARIOS

Objective: To prepare positive test cases for the main functions of a hospital appointment booking system.

Case Study: The system allows patients to create accounts, search doctors by speciality, view available slots, book appointments, receive confirmations and review appointment history.

Requirement Analysis:

- REQ-01: Patients shall register and sign in using valid details.
- REQ-02: Doctors shall be searchable by speciality and location.
- REQ-03: Patients shall book an available appointment slot.
- REQ-04: The system shall issue an appointment reference after booking.
- REQ-05: Patients shall view their upcoming and completed appointments.

Features to be tested:

1. Patient registration

2. Patient login

3. Doctor search

4. Slot selection

5. Appointment booking

6. Confirmation

7. Appointment history

Test Case ID: HOSP_P01
Test Description: Register a patient with valid details

Steps to reproduce:

40.1. Open the patient portal.
41.2. Enter name, contact details and required information.
42.3. Select Register.

Expected Result: A patient account is created and a confirmation is displayed.

Actual Result: A patient account is created and a confirmation is displayed.

Status: Pass

Test Case ID: HOSP_P02
Test Description: Log in with valid patient credentials

Steps to reproduce:

43.1. Open the patient login page.
44.2. Enter the registered credentials.
45.3. Select Login.

Expected Result: The patient dashboard is displayed.

Actual Result: The patient dashboard is displayed.

Status: Pass

Test Case ID: HOSP_P03
Test Description: Search for a doctor by speciality

Steps to reproduce:

46.1. Open Find a Doctor.
47.2. Select Cardiology.
48.3. Run the search.

Expected Result: Doctors matching the selected speciality are listed.

Actual Result: Doctors matching the selected speciality are listed.

Status: Pass

Test Case ID: HOSP_P04
Test Description: Book an available appointment slot

Steps to reproduce:

49.1. Select a doctor.
50.2. Choose an available date and time.
51.3. Confirm the appointment.

Expected Result: The appointment is saved with the selected date and time.

Actual Result: The appointment is saved with the selected date and time.

Status: Pass

Test Case ID: HOSP_P05
Test Description: View appointment confirmation

Steps to reproduce:

52.1. Complete a valid booking.
53.2. Open appointment details.

Expected Result: A unique appointment reference and booking details are displayed.

Actual Result: A unique appointment reference and booking details are displayed.

Status: Pass

Test Case ID: HOSP_P06
Test Description: View upcoming appointments

Steps to reproduce:

54.1. Sign in to the patient portal.
55.2. Open My Appointments.

Expected Result: All upcoming appointments for the patient are listed accurately.

Actual Result: All upcoming appointments for the patient are listed accurately.

Status: Pass

Test Case ID: HOSP_P07
Test Description: Download an appointment receipt

Steps to reproduce:

56.1. Open a confirmed appointment.
57.2. Select Download Receipt.

Expected Result: A receipt containing the correct patient and appointment details is downloaded.

Actual Result: A receipt containing the correct patient and appointment details is downloaded.

Status: Pass

Result: 7 of 7 test cases passed.

EXPERIMENT NO: 9

HOSPITAL APPOINTMENT SYSTEM - NEGATIVE TEST SCENARIOS

Objective: To test whether invalid appointment requests are rejected safely and consistently.

Case Study: Negative testing verifies missing patient information, invalid dates, unavailable slots, duplicate bookings and unauthorised access to appointment details.

Requirement Analysis:

- REQ-01: Mandatory patient fields shall be validated.
- REQ-02: Past dates shall not be accepted for new appointments.
- REQ-03: A slot already booked by another patient shall not be booked again.
- REQ-04: Duplicate bookings for the same slot shall be prevented.
- REQ-05: Patients shall access only their own appointment records.

Features to be tested:

1. Required-field validation

2. Date validation

3. Slot availability

4. Duplicate booking

5. Invalid contact details

6. Unauthorised access

7. Session expiry

Test Case ID: HOSP_N01
Test Description: Register with a missing phone number

Steps to reproduce:

58.1. Open the registration form.
59.2. Leave the phone number blank.
60.3. Select Register.

Expected Result: A validation message is displayed and the account is not created.

Actual Result: A validation message is displayed and the account is not created.

Status: Pass

Test Case ID: HOSP_N02
Test Description: Choose a date in the past

Steps to reproduce:

61.1. Open the appointment booking page.
62.2. Select a past date.
63.3. Try to continue.

Expected Result: The date is rejected and a valid-date message is displayed.

Actual Result: The date is rejected and a valid-date message is displayed.

Status: Pass

Test Case ID: HOSP_N03
Test Description: Book a slot that is no longer available

Steps to reproduce:

64.1. Open a doctor schedule.
65.2. Select a slot that has already been taken.
66.3. Confirm the booking.

Expected Result: The booking is rejected and the user is asked to select another slot.

Actual Result: The booking is rejected and the user is asked to select another slot.

Status: Pass

Test Case ID: HOSP_N04
Test Description: Book the same appointment twice

Steps to reproduce:

67.1. Book a slot successfully.
68.2. Repeat the same booking request.

Expected Result: The system prevents a duplicate appointment.

Actual Result: The system prevents a duplicate appointment.

Status: Pass

Test Case ID: HOSP_N05
Test Description: Enter an invalid email address

Steps to reproduce:

69.1. Open the patient registration form.
70.2. Enter an incorrectly formatted email.
71.3. Submit the form.

Expected Result: An email validation message is displayed.

Actual Result: An email validation message is displayed.

Status: Pass

Test Case ID: HOSP_N06
Test Description: Access another patient's appointment

Steps to reproduce:

72.1. Sign in as one patient.
73.2. Try to open an appointment reference belonging to another patient.

Expected Result: Access is denied and private appointment information is not shown.

Actual Result: Access is denied and private appointment information is not shown.

Status: Pass

Test Case ID: HOSP_N07
Test Description: Use the portal after session expiry

Steps to reproduce:

74.1. Sign in to the portal.
75.2. Remain inactive until the session expires.
76.3. Try to open appointment details.

Expected Result: The user is redirected to login and protected information remains unavailable.

Actual Result: The user is redirected to login and protected information remains unavailable.

Status: Pass

Result: 7 of 7 test cases passed.

EXPERIMENT NO: 10

TEST PLAN - HOTEL RESERVATION SYSTEM

Objective: To prepare a test plan defining the scope, strategy, resources, schedule and completion criteria for a Hotel Reservation System.

1. Introduction

The Hotel Reservation System supports room search, room availability checks, guest registration, booking, payment, cancellation and invoice generation. This plan describes the approach used to verify that reservations are accurate, secure and reliable.

2. Test Scope

In scope: Guest login and registration; hotel and room search; date and guest-count validation; room availability; reservation creation; payment confirmation; booking cancellation; invoice generation.

Out of scope: Physical hotel operations, external bank infrastructure, third-party map services and operating-system internals.

3. Test Strategy and Levels of Testing

- Unit Testing: Validate individual search, pricing and booking functions.
- Integration Testing: Verify data flow between availability, reservation and payment modules.
- System Testing: Test complete booking workflows in a QA environment.
- Acceptance Testing: Hotel staff validate common guest journeys.
- Regression Testing: Re-run existing tests after fixes or changes.
- Non-functional Testing: Evaluate response time, accessibility, privacy and concurrent bookings.

4. Test Design Techniques

- Equivalence Partitioning for guest counts and date inputs.
- Boundary Value Analysis for minimum/maximum stay and booking dates.
- Decision Table Testing for room eligibility and cancellation fees.
- State Transition Testing for reservation states: Pending, Confirmed, Cancelled and Checked-in.
- Use Case Testing for search-to-checkout workflows.
- Error Guessing for interrupted payment and repeated booking submissions.

5. Test Environment

- Hardware: Quad-core processor, 8 GB RAM and SSD.
- Operating System: Windows 11 or Ubuntu Linux.
- Browsers: Current Chrome, Firefox and Edge.
- Application and database: QA deployment with a relational database.
- Tools: Selenium for browser automation, Postman for API checks, JIRA for defect tracking and a spreadsheet for test data.

6. Roles and Responsibilities

- Test Lead: Approves the plan, tracks progress and prepares the summary.
- QA Engineers: Design and execute test cases and report defects.
- Developers: Investigate and fix reported issues.
- Hotel Representative: Performs acceptance testing and confirms business rules.

7. Entry Criteria

- Requirements and acceptance rules are approved.
- The test build and environment are available.
- Test cases have been reviewed.
- Valid, invalid and boundary test data are prepared.

8. Test Execution Strategy

- Run smoke tests on each new build.
- Execute critical booking and payment tests first.
- Record each failure with steps, evidence, severity and priority.
- Retest fixes and run regression tests before release.

9. Suspension Criteria

- The build prevents execution of a major part of the test suite.
- The QA environment or database is unavailable.
- Payment simulation is not functioning and blocks required test cases.

10. Exit / Test Completion Criteria

- All planned high-priority test cases have been executed.
- At least 95% of executed test cases pass.
- No unresolved critical defects remain.
- Reservation and payment records are consistent.
- The test summary report is reviewed and approved.

11. Risks and Mitigation

- Concurrent bookings may assign the same room: test simultaneous requests and verify transactional locking.
- Changes to cancellation rules may invalidate tests: maintain requirement traceability and review test cases.
- Payment sandbox downtime may delay testing: prepare a documented mock payment option.
- Insufficient test data may hide date-related defects: create cases covering weekdays, weekends and date boundaries.

12. Test Schedule

- Planning and estimation: 2 days.
- Test design and review: 3 days.
- Environment and test-data setup: 2 days.
- Test execution: 5 days.
- Defect retesting and regression: 3 days.
- Closure and reporting: 1 day.

13. Test Deliverables

- Test Plan document.
- Test scenarios and test cases.
- Requirement Traceability Matrix.
- Defect log.
- Execution summary and final test report.

14. Sample Test Cases and Execution Status

Test Case ID: HOTEL_TC01
Module: Login

Test Description: Guest login with valid credentials

Expected Result: Guest dashboard is displayed.

Actual Result: Guest dashboard is displayed.

Status: Pass

Test Case ID: HOTEL_TC02
Module: Search

Test Description: Search rooms for a valid date range

Expected Result: Matching available rooms are listed.

Actual Result: Matching available rooms are listed.

Status: Pass

Test Case ID: HOTEL_TC03
Module: Availability

Test Description: Search when checkout date is earlier than check-in

Expected Result: Date validation message is displayed.

Actual Result: Date validation message is displayed.

Status: Pass

Test Case ID: HOTEL_TC04
Module: Reservation

Test Description: Reserve an available room with valid guest details

Expected Result: Reservation is created with a reference number.

Actual Result: Reservation is created with a reference number.

Status: Pass

Test Case ID: HOTEL_TC05
Module: Payment

Test Description: Complete payment using a valid test method

Expected Result: Payment confirmation is recorded.

Actual Result: Payment confirmation is recorded.

Status: Pass

Test Case ID: HOTEL_TC06
Module: Cancellation

Test Description: Cancel a refundable reservation

Expected Result: Reservation is cancelled and refund information is shown.

Actual Result: Reservation is cancelled and refund information is shown.

Status: Pass

Test Case ID: HOTEL_TC07
Module: Invoice

Test Description: Download invoice for a confirmed booking

Expected Result: Invoice displays the correct guest, dates and amount.

Actual Result: Invoice displays the correct guest, dates and amount.

Status: Pass

Result: 7 of 7 test cases passed.

EXPERIMENT NO: 11

BUS TICKET BOOKING SYSTEM - POSITIVE TEST SCENARIOS

Objective: To prepare positive test cases for the important functions of an online bus ticket booking system.

Case Study: Passengers can search routes, select a bus and seat, enter traveller details, pay the fare, receive a ticket and view booking information.

Requirement Analysis:

- REQ-01: Users shall search trips using valid origin, destination and date.
- REQ-02: Available buses and seats shall be displayed.
- REQ-03: Valid passenger details shall be accepted.
- REQ-04: Successful payment shall generate a booking reference and ticket.
- REQ-05: Users shall view ticket details after booking.

Features to be tested:

1. Route search

2. Bus listing

3. Seat selection

4. Passenger details

5. Payment

6. Ticket generation

7. Booking history

Test Case ID: BUS_P01
Test Description: Search buses for a valid route

Steps to reproduce:

77.1. Open the bus booking site.
78.2. Enter origin, destination and a future date.
79.3. Select Search Buses.

Expected Result: Available buses with timings and fares are displayed.

Actual Result: Available buses with timings and fares are displayed.

Status: Pass

Test Case ID: BUS_P02
Test Description: View seat layout

Steps to reproduce:

80.1. Choose a listed bus.
81.2. Open seat selection.

Expected Result: Available and booked seats are shown with a clear legend.

Actual Result: Available and booked seats are shown with a clear legend.

Status: Pass

Test Case ID: BUS_P03
Test Description: Select an available seat

Steps to reproduce:

82.1. Choose an available seat.
83.2. Continue to passenger details.

Expected Result: The selected seat is reserved temporarily for the booking session.

Actual Result: The selected seat is reserved temporarily for the booking session.

Status: Pass

Test Case ID: BUS_P04
Test Description: Enter valid passenger details

Steps to reproduce:

84.1. Enter passenger name, age and contact details.
85.2. Continue to checkout.

Expected Result: Valid details are accepted and the fare summary is displayed.

Actual Result: Valid details are accepted and the fare summary is displayed.

Status: Pass

Test Case ID: BUS_P05
Test Description: Pay the fare successfully

Steps to reproduce:

86.1. Select a supported payment method.
87.2. Complete the payment using valid test details.

Expected Result: Payment succeeds and a booking reference is generated.

Actual Result: Payment succeeds and a booking reference is generated.

Status: Pass

Test Case ID: BUS_P06
Test Description: View the booked ticket

Steps to reproduce:

88.1. Open My Bookings.
89.2. Select the new booking.

Expected Result: The ticket shows the correct route, travel date, seat and passenger details.

Actual Result: The ticket shows the correct route, travel date, seat and passenger details.

Status: Pass

Test Case ID: BUS_P07
Test Description: Download the e-ticket

Steps to reproduce:

90.1. Open a confirmed booking.
91.2. Select Download Ticket.

Expected Result: The e-ticket is downloaded successfully.

Actual Result: The e-ticket is downloaded successfully.

Status: Pass

Result: 7 of 7 test cases passed.

EXPERIMENT NO: 12

BUS TICKET BOOKING SYSTEM - NEGATIVE TEST SCENARIOS

Objective: To ensure invalid bus booking requests are rejected and seat or payment records remain correct.

Case Study: Negative tests cover invalid routes, past travel dates, unavailable seats, invalid passenger information, failed payments and invalid booking references.

Requirement Analysis:

- REQ-01: Origin and destination must be valid and different.
- REQ-02: Travel dates in the past shall be rejected.
- REQ-03: A seat already sold shall not be assigned again.
- REQ-04: Invalid passenger data shall be rejected.
- REQ-05: Failed payment shall not confirm a ticket.

Features to be tested:

1. Route validation

2. Date validation

3. Seat conflict

4. Passenger validation

5. Contact validation

6. Payment failure

7. Booking reference validation

Test Case ID: BUS_N01
Test Description: Search with identical origin and destination

Steps to reproduce:

92.1. Enter the same city in both route fields.
93.2. Select Search Buses.

Expected Result: A message asks the user to choose different origin and destination locations.

Actual Result: A message asks the user to choose different origin and destination locations.

Status: Pass

Test Case ID: BUS_N02
Test Description: Select a past travel date

Steps to reproduce:

94.1. Enter a valid route.
95.2. Choose a date in the past.
96.3. Search for buses.

Expected Result: The system rejects the date and displays a validation message.

Actual Result: The system rejects the date and displays a validation message.

Status: Pass

Test Case ID: BUS_N03
Test Description: Select a seat that has been sold

Steps to reproduce:

97.1. Open a bus seat layout.
98.2. Attempt to select a seat marked unavailable.

Expected Result: The unavailable seat cannot be selected.

Actual Result: The unavailable seat cannot be selected.

Status: Pass

Test Case ID: BUS_N04
Test Description: Continue with a blank passenger name

Steps to reproduce:

99.1. Select an available seat.
100. 2. Leave the passenger name blank.
101. 3. Continue to payment.

Expected Result: A required-field message appears and checkout is blocked.

Actual Result: A required-field message appears and checkout is blocked.

Status: Pass

Test Case ID: BUS_N05
Test Description: Enter an invalid contact number

Steps to reproduce:

102. 1. Open passenger details.
103. 2. Enter a malformed phone number.
104. 3. Submit the details.

Expected Result: The contact number is rejected with a clear validation message.

Actual Result: The contact number is rejected with a clear validation message.

Status: Pass

Test Case ID: BUS_N06
Test Description: Complete booking after payment is declined

Steps to reproduce:

105. 1. Proceed to checkout.
106. 2. Use a declined test payment method.

Expected Result: The ticket remains unconfirmed and no successful payment is recorded.

Actual Result: The ticket remains unconfirmed and no successful payment is recorded.

Status: Pass

Test Case ID: BUS_N07
Test Description: Search using an invalid booking reference

Steps to reproduce:

107. 1. Open booking lookup.
108. 2. Enter a malformed reference number.
109. 3. Select Search.

Expected Result: An invalid-reference message is displayed without exposing other bookings.

Actual Result: An invalid-reference message is displayed without exposing other bookings.

Status: Pass

Result: 7 of 7 test cases passed.

EXPERIMENT NO: 13

STUDENT ATTENDANCE SYSTEM - POSITIVE TEST SCENARIOS

Objective: To verify the main operations of a student attendance management system through positive test cases.

Case Study: The system allows faculty to log in, select a class, mark student attendance, save records and generate attendance summaries for a selected period.

Requirement Analysis:

- REQ-01: Faculty shall sign in with valid credentials.
- REQ-02: Faculty shall view assigned classes and student lists.
- REQ-03: Attendance shall be saved for a valid class and date.
- REQ-04: The system shall calculate attendance percentages correctly.
- REQ-05: Reports shall be generated for the selected class and period.

Features to be tested:

1. Faculty login

2. Class list

3. Student roster

4. Attendance entry

5. Save attendance

6. Percentage calculation

7. Report generation

Test Case ID: ATT_P01
Test Description: Faculty login with valid credentials

Steps to reproduce:

110. 1. Open the attendance portal.
111. 2. Enter valid faculty credentials.
112. 3. Select Login.

Expected Result: The faculty dashboard is displayed.

Actual Result: The faculty dashboard is displayed.

Status: Pass

Test Case ID: ATT_P02
Test Description: Open an assigned class

Steps to reproduce:

113. 1. Sign in as faculty.
114. 2. Select My Classes.
115. 3. Open an assigned class.

Expected Result: The class details and student roster are displayed.

Actual Result: The class details and student roster are displayed.

Status: Pass

Test Case ID: ATT_P03
Test Description: Mark a student present

Steps to reproduce:

116. 1. Open the class attendance page.
117. 2. Mark a student as Present.
118. 3. Save attendance.

Expected Result: The student's attendance status is saved.

Actual Result: The student's attendance status is saved.

Status: Pass

Test Case ID: ATT_P04
Test Description: Mark absent students

Steps to reproduce:

119. 1. Open the roster.
120. 2. Mark selected students as Absent.
121. 3. Save the session.

Expected Result: The absent statuses are saved for the selected date.

Actual Result: The absent statuses are saved for the selected date.

Status: Pass

Test Case ID: ATT_P05
Test Description: Edit attendance before final submission

Steps to reproduce:

122. 1. Open an editable attendance session.
123. 2. Change a student's status.
124. 3. Save the update.

Expected Result: The latest status is stored and displayed.

Actual Result: The latest status is stored and displayed.

Status: Pass

Test Case ID: ATT_P06
Test Description: Calculate attendance percentage

Steps to reproduce:

125. 1. Open a student's attendance summary.
126. 2. Select the semester period.

Expected Result: The attendance percentage matches the saved present and total session counts.

Actual Result: The attendance percentage matches the saved present and total session counts.

Status: Pass

Test Case ID: ATT_P07
Test Description: Generate a class attendance report

Steps to reproduce:

127. 1. Select a class and date range.
128. 2. Choose Generate Report.

Expected Result: A report is generated with the correct attendance records.

Actual Result: A report is generated with the correct attendance records.

Status: Pass

Result: 7 of 7 test cases passed.

EXPERIMENT NO: 14

STUDENT ATTENDANCE SYSTEM - NEGATIVE TEST SCENARIOS

Objective: To verify that invalid attendance data and unauthorised changes are prevented.

Case Study: Negative testing checks invalid login attempts, missing class selection, duplicate submissions, invalid dates and access by users who are not assigned to a class.

Requirement Analysis:

- REQ-01: Invalid faculty credentials shall be rejected.
- REQ-02: Attendance cannot be saved without a class and date.
- REQ-03: Duplicate attendance sessions for the same class and date shall be prevented.
- REQ-04: Attendance dates outside permitted rules shall be rejected.
- REQ-05: Only authorised faculty may edit class attendance.

Features to be tested:

1. Login error handling

2. Required fields

3. Duplicate session prevention

4. Date validation

5. Student roster validation

6. Role-based access

7. Edit permissions

Test Case ID: ATT_N01
Test Description: Login with an incorrect password

Steps to reproduce:

129. 1. Open the faculty login page.
130. 2. Enter a valid user ID and incorrect password.
131. 3. Select Login.

Expected Result: An invalid-credentials message appears and access is denied.

Actual Result: An invalid-credentials message appears and access is denied.

Status: Pass

Test Case ID: ATT_N02
Test Description: Save attendance without selecting a class

Steps to reproduce:

132. 1. Open the attendance form.
133. 2. Leave the class field empty.
134. 3. Select Save.

Expected Result: The system requests a class selection and does not save the record.

Actual Result: The system requests a class selection and does not save the record.

Status: Pass

Test Case ID: ATT_N03
Test Description: Save attendance without a date

Steps to reproduce:

135. 1. Select a class.
136. 2. Leave the date field empty.
137. 3. Select Save.

Expected Result: A date validation message appears.

Actual Result: A date validation message appears.

Status: Pass

Test Case ID: ATT_N04
Test Description: Create duplicate attendance for the same session

Steps to reproduce:

138. 1. Save attendance for a class and date.
139. 2. Try to create another session for the same class and date.

Expected Result: The duplicate session is blocked or routed to the existing record.

Actual Result: The duplicate session is blocked or routed to the existing record.

Status: Pass

Test Case ID: ATT_N05
Test Description: Enter a date outside the permitted period

Steps to reproduce:

140. 1. Open attendance entry.
141. 2. Choose a date outside the allowed editing period.
142. 3. Attempt to save.

Expected Result: The system rejects the date and explains the restriction.

Actual Result: The system rejects the date and explains the restriction.

Status: Pass

Test Case ID: ATT_N06
Test Description: Edit a class not assigned to the faculty member

Steps to reproduce:

143. 1. Sign in as faculty.
144. 2. Attempt to open another faculty member's class attendance.

Expected Result: Access is denied and the attendance record is unchanged.

Actual Result: Access is denied and the attendance record is unchanged.

Status: Pass

Test Case ID: ATT_N07
Test Description: Submit attendance with a missing student status

Steps to reproduce:

145. 1. Open a class roster.
146. 2. Leave one student's status unselected.
147. 3. Select Submit.

Expected Result: The system highlights the missing status and prevents incomplete submission.

Actual Result: The system highlights the missing status and prevents incomplete submission.

Status: Pass

Result: 7 of 7 test cases passed.

EXPERIMENT NO: 15

TEST PLAN - VEHICLE RENTAL MANAGEMENT SYSTEM

Objective: To develop a test plan for a Vehicle Rental Management System covering scope, test levels, test environment, execution and completion criteria.

1. Introduction

The Vehicle Rental Management System manages vehicle listings, availability, customer registration, booking, rental charges, pickup, return and invoices. The test plan defines how the system will be checked for functional correctness, data integrity and secure access.

2. Test Scope

In scope: Customer registration and login; vehicle search and filters; availability by date; booking creation; rental price calculation; pickup and return updates; cancellation; invoice generation; admin vehicle management.

Out of scope: Vehicle hardware diagnostics, external insurance-provider systems, payment gateway internals and physical vehicle maintenance.

3. Test Strategy and Levels of Testing

- Unit Testing: Verify price calculation, date validation and availability functions.
- Integration Testing: Check links between vehicle inventory, booking, payment and invoice modules.
- System Testing: Execute end-to-end rental scenarios in the QA environment.
- Acceptance Testing: Rental desk staff validate business workflows.
- Regression Testing: Re-run critical tests after code changes.
- Non-functional Testing: Assess usability, access control, response time and simultaneous booking behaviour.

4. Test Design Techniques

- Equivalence Partitioning for age, rental duration and vehicle category.
- Boundary Value Analysis for minimum rental period and booking date limits.
- Decision Table Testing for eligibility, deposit and cancellation rules.
- State Transition Testing for vehicle states: Available, Reserved, Rented, Under Maintenance.
- Use Case Testing for booking, pickup and return.

- Error Guessing for duplicate bookings, delayed payment responses and missed returns.

5. Test Environment

- Hardware: Standard test workstation with 8 GB RAM.
- Operating System: Windows 11 or Ubuntu Linux.
- Browsers: Current Chrome, Firefox and Edge.
- Application: QA instance connected to a test database and payment sandbox.
- Tools: Selenium, Postman, JIRA and a spreadsheet for test data.

6. Roles and Responsibilities

- Test Lead: Plans testing and reviews completion status.
- QA Testers: Prepare test data, execute test cases and report defects.
- Developers: Resolve defects and support technical analysis.
- Rental Operations Representative: Performs acceptance testing.

7. Entry Criteria

- Approved requirements and rental policies are available.
- The QA build is deployed and smoke tests pass.
- Test cases are reviewed and test accounts are created.
- Vehicle, date, pricing and customer test data are prepared.

8. Test Execution Strategy

- Execute smoke tests after each deployment.
- Prioritise booking, availability and payment workflows.
- Record failures with evidence, severity and reproducible steps.
- Retest corrected defects and run regression tests before sign-off.

9. Suspension Criteria

- The QA environment is unavailable.
- A critical defect prevents most booking tests from running.
- The test payment service is unavailable and no approved simulation is available.

10. Exit / Test Completion Criteria

- All critical test cases have been executed.
- At least 95% of executed cases pass.
- No critical or high-severity defects remain open.
- Vehicle availability and rental charges are consistent across modules.

- The final test report is approved by the test lead.

11. Risks and Mitigation

- Two customers may reserve the same vehicle: test concurrent requests and enforce atomic reservation updates.
- Incorrect rental charges may be calculated: test date boundaries, extensions and discounts.
- Late policy changes may affect expected results: maintain traceability and review affected cases.
- QA payment failures may delay testing: maintain a documented payment simulator.

12. Test Schedule

- Planning: 2 days.
- Test design and review: 4 days.
- Environment and data setup: 2 days.
- Execution: 6 days.
- Retesting and regression: 3 days.
- Closure report: 1 day.

13. Test Deliverables

- Test Plan.
- Test scenarios and test cases.
- Requirement Traceability Matrix.
- Defect log.
- Test execution and summary report.

14. Sample Test Cases and Execution Status

Test Case ID: VRM_TC01
Module: Login

Test Description: Customer login with valid credentials

Expected Result: Customer dashboard is displayed.

Actual Result: Customer dashboard is displayed.

Status: Pass

Test Case ID: VRM_TC02
Module: Vehicle Search

Test Description: Search for an available vehicle by category

Expected Result: Matching vehicles are listed.

Actual Result: Matching vehicles are listed.

Status: Pass

Test Case ID: VRM_TC03
Module: Availability

Test Description: Search for a vehicle for a valid date range

Expected Result: Availability is displayed for the selected dates.

Actual Result: Availability is displayed for the selected dates.

Status: Pass

Test Case ID: VRM_TC04
Module: Booking

Test Description: Book an available vehicle with valid details

Expected Result: Booking reference is generated.

Actual Result: Booking reference is generated.

Status: Pass

Test Case ID: VRM_TC05
Module: Pricing

Test Description: Calculate charges for a multi-day rental

Expected Result: Total charge matches the configured daily rate and fees.

Actual Result: Total charge matches the configured daily rate and fees.

Status: Pass

Test Case ID: VRM_TC06
Module: Return

Test Description: Record the return of a rented vehicle

Expected Result: Vehicle status changes to Available after required checks.

Actual Result: Vehicle status changes to Available after required checks.

Status: Pass

Test Case ID: VRM_TC07
Module: Invoice

Test Description: Generate invoice for a completed rental

Expected Result: Invoice displays the correct dates and charges.

Actual Result: Invoice displays the correct dates and charges.

Status: Pass

Result: 7 of 7 test cases passed.